

# Packet Tracer Lab 20 - Troubleshoot a Wireless Connection

## Objectives

Identify and correct any misconfiguration of a wireless device.

## Background / Scenario

A small business owner learns that a wireless user is unable to access the network. All the PCs are configured with static IP addressing. Identify and resolve the issue.

## Instructions

### Part 1: Verify connectivity.

Access the **Desktop > Web Browser** of each wireless PC and type **www.cisco.pka** into the URL. Identify any PCs that are not connecting to the web server.

**Note:** All the devices require time to complete the boot process. Please allow up to one minute before receiving a web response.

Which wireless PCs are unable to connect to the web server? **No PCs can access the web server, "host name unresolved"**

### Part 2: Examine the IP configuration of the PCs.

- On the PC that is unable to connect, access the **Command Prompt** from the **Desktop** tab.
- Type the **ipconfig /all** command.

What IP addressing information is available? **PC1 is using the Addr/mask 192.254.122.96/16**

### Part 3: Examine the Wireless Settings on the Wireless Client.

- On the **Desktop** tab of any PC that is unable to connect, click **PC Wireless** to access the wireless configurations.
- Click **Connect** tab and record the associated SSID. Click **Refresh** as needed to display the list of SSID. **Academy**

### Part 4: Examine the Wireless Settings on the Wireless Router.

- Access the Wireless Router from the web browser of a wired PC. Use the username **admin** and password **admin** to access the wireless router.

What IP address did you use? **192.168.1.1** (Hint: default gateway)

- On the **Basic Setup** page, examine the **DHCP Server Setting** configuration.

Is DHCP enabled? **Yes**

- d. Click the **Wireless** tab.
- e. Examine the setup information under the **Wireless** tab.

What is the SSID? **Academy** Does it match the SSID configured on the client? **Yes**

- f. Click the **Wireless Security** submenu.
- g. Examine the security settings.

What is the wireless security mode? **WPA2 Personal** What is the passphrase?  
**Cisco123**

## **Part 5: Make any necessary configuration changes on the Wireless Clients**

- a. On the **Desktop** tab of any PC that is unable to connect. Click **PC Wireless** to correct the wireless configurations.
- b. Click **Connect** tab. Select the Academy wireless network and click **Connect**.
- c. Enter the passphrase (Pre-shared Key) recorded from the wireless router. Click **Connect**.
- d. Using the **Web Browser** within the **Desktop** tab connect to **www.cisco.pka** to verify that the configuration changes resolved the problem.